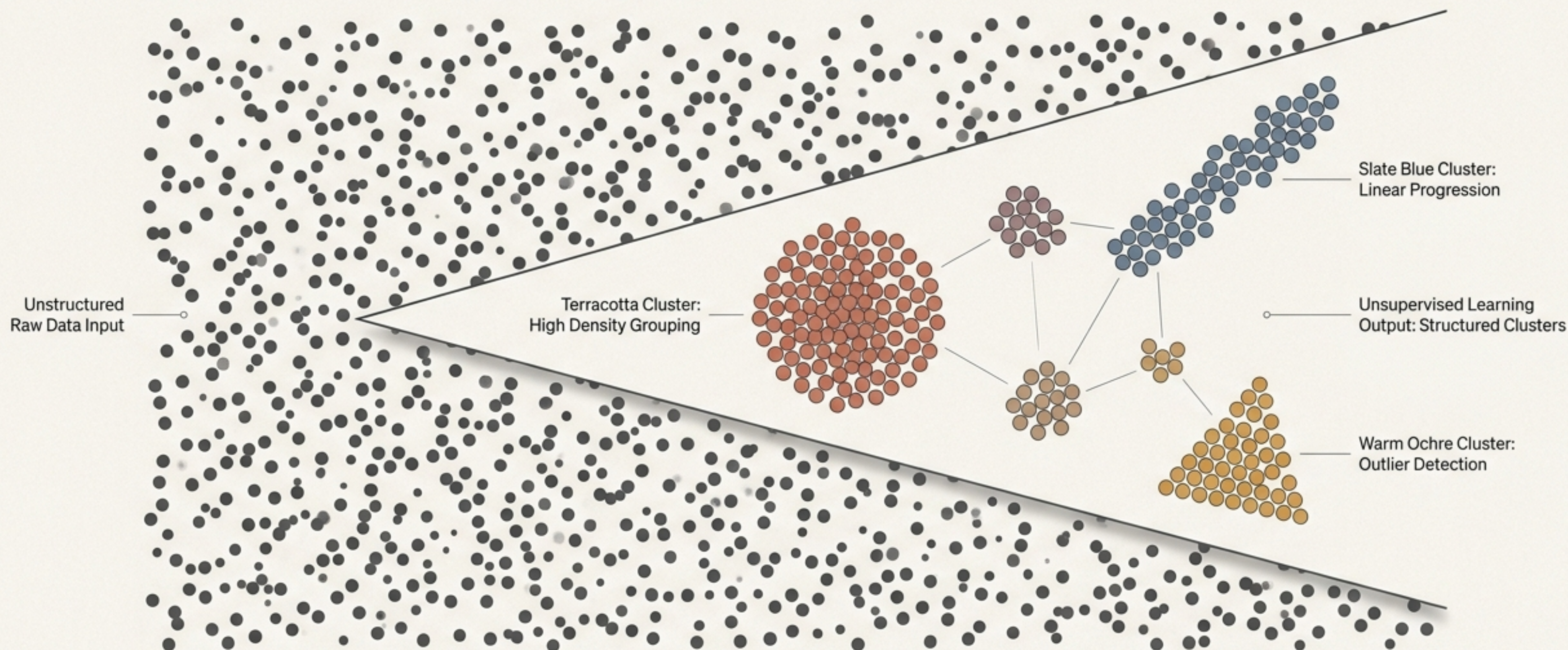


Illuminating the Invisible

A Practitioner's Playbook for Unsupervised Learning

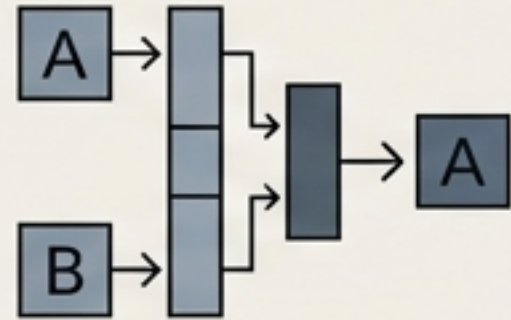


```
model = KMeans(n_clusters=3); model.fit(data); labels = model.predict(data)
```

```
algorithm: t-SNE / DBSCAN
```

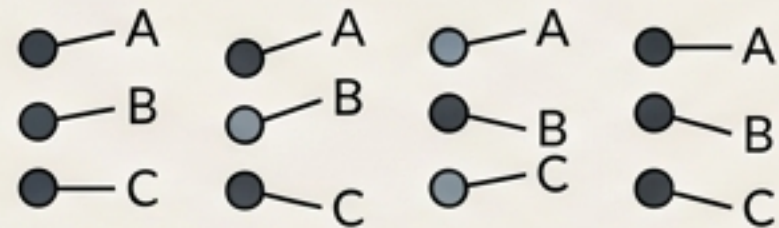


Mapping the machine learning universe



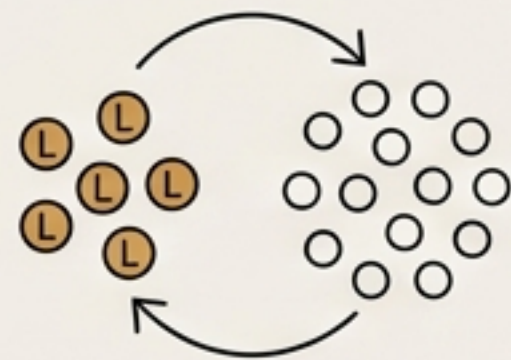
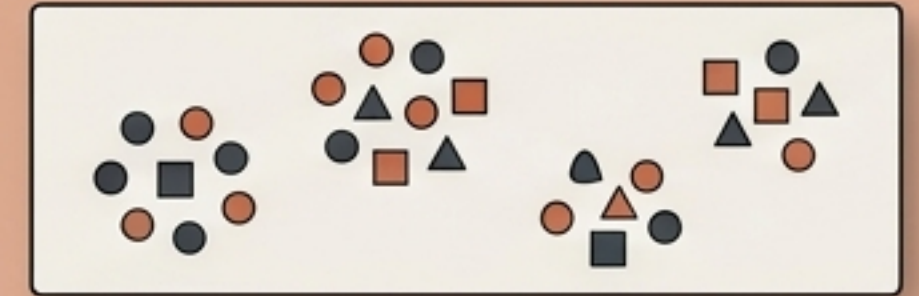
Supervised Learning

Trains on labeled data to predict unknown labels. Examples: Classification, Regression.



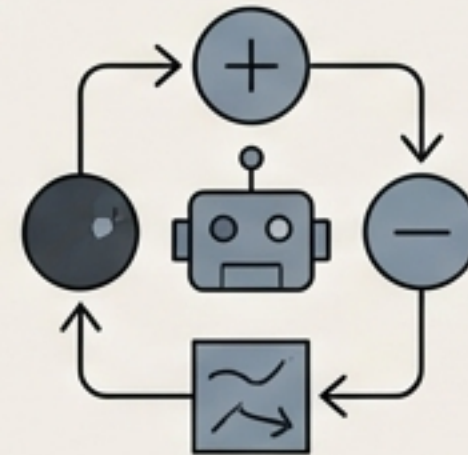
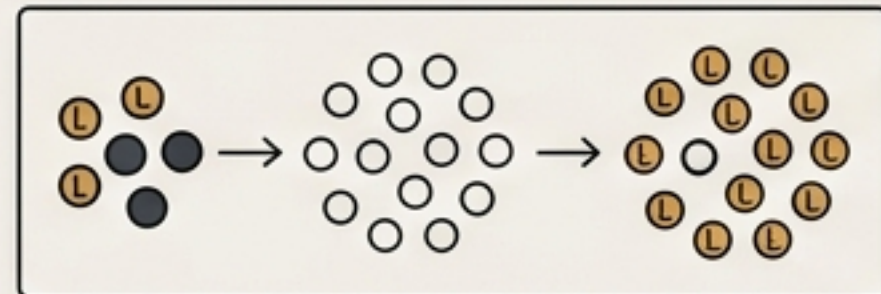
Unsupervised Learning

Finds patterns and hidden structures in purely unlabeled data.



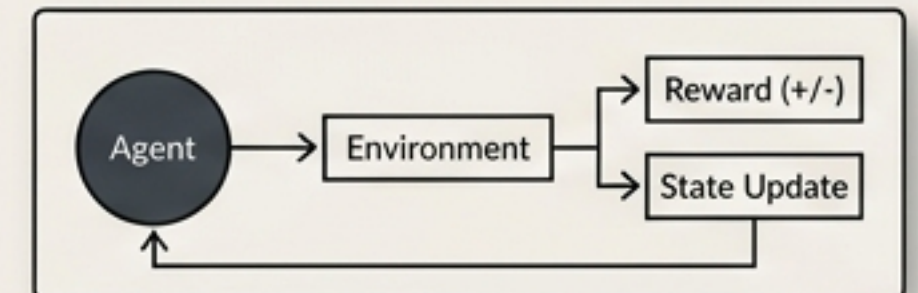
Semi-Supervised Learning

Trains on a small subset of labeled data and iteratively labels the rest.



Reinforcement Learning

An agent learning iteratively from feedback within an environment.

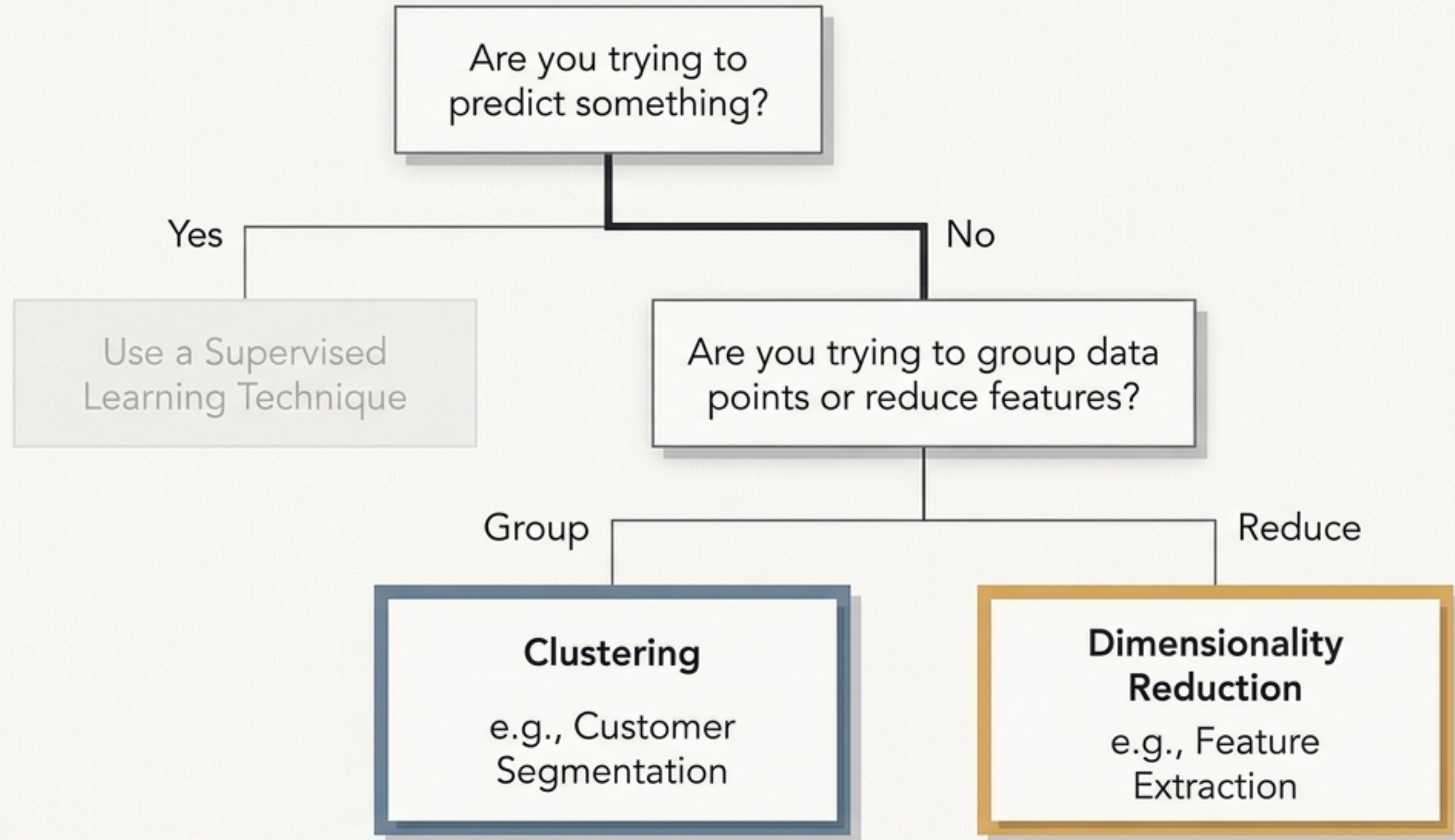


Finding structure without labels requires a fundamental shift in analytical mindset

	Supervised	Unsupervised
Data Type	Labeled data pairs	Unlabeled raw data
Core Goal	Predict specific outcomes	Describe, organize, and discover patterns
Outputs	Categorical classes or continuous numerical values	Clusters, structures, and reduced dimensions
Evaluation	Mathematical accuracy metrics	Domain expertise, intuition, and business objective fulfillment



The unsupervised decision matrix

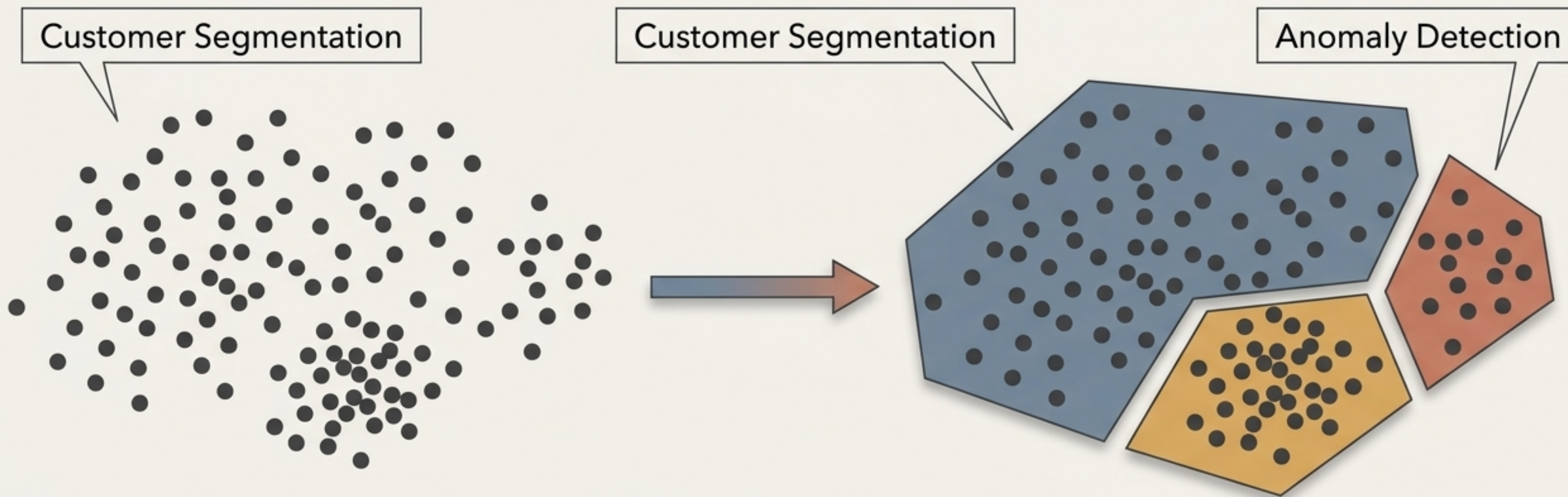


Core Branch 1: The art of grouping and segmentation

Clustering partitions unlabeled data based on underlying similarities. Evaluation relies on heuristic methods like silhouette analysis, the elbow method, and the Davies-Bouldin Index.

```
# Evaluation Metrics
from sklearn.metrics import silhouette_score, davies_bouldin_score
from yellowbrick.cluster import KElbowVisualizer

silhouette_avg = silhouette_score(X, labels)
davies_bouldin = davies_bouldin_score(X, labels)
visualizer = KElbowVisualizer(model, k=(2,10))
```



Clustering playbook: Centroid and hierarchical methods

K-Means Clustering

Mechanics: Partitions data based on distance between data points and central centroids.

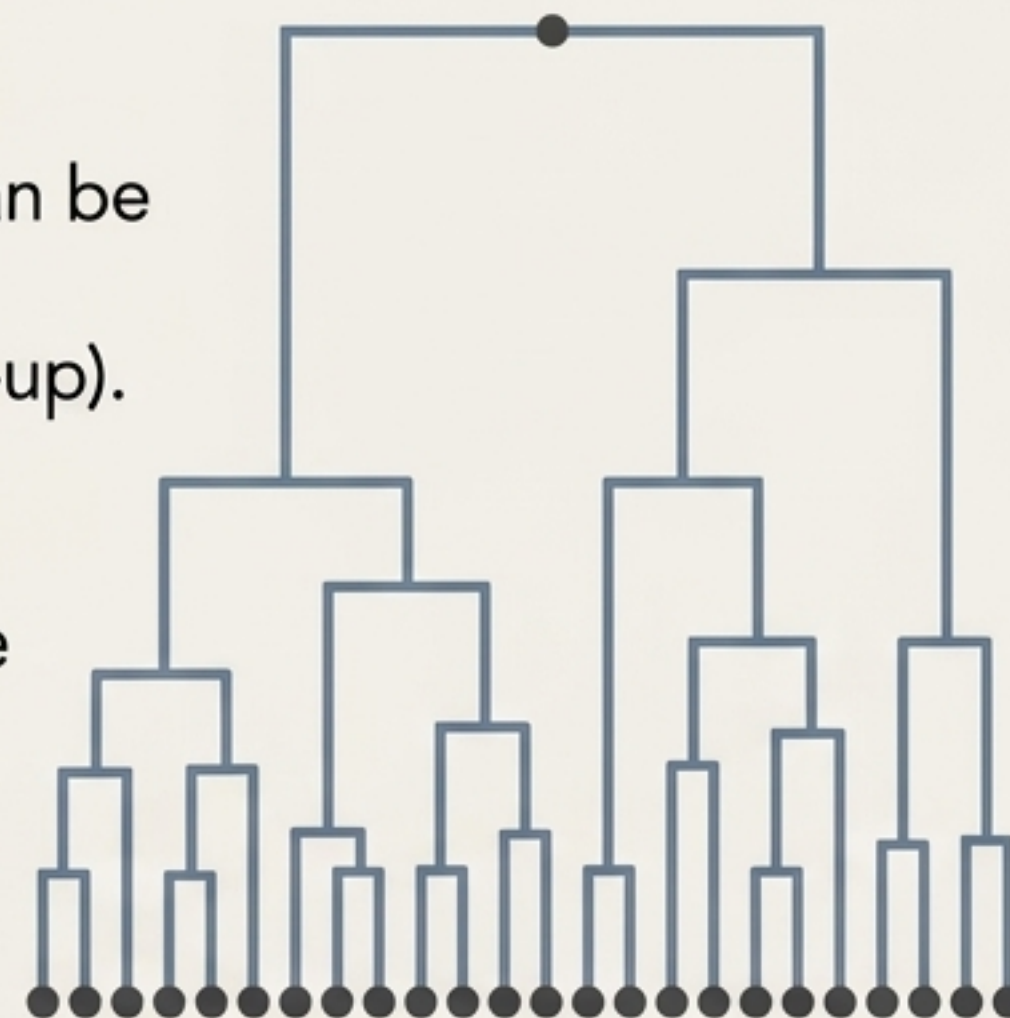
Strengths: Fast, highly interpretable.

Weaknesses: Struggles with imbalanced cluster sizes or non-convex (irregular) shapes.

Hierarchical Clustering

Mechanics: Builds a hierarchy of clusters. Can be divisive (top-down) or agglomerative (bottom-up).

Output: Produces a dendrogram to visualize the cluster hierarchy.



Clustering playbook: Density-based methods

DBSCAN

Density-Based Spatial Clustering of Applications with Noise

Mechanics: Creates clusters based on density; works perfectly with natural, irregular patterns.

Pros/Cons: Identifies outliers automatically, no predefined cluster count required. Struggles with varying densities.

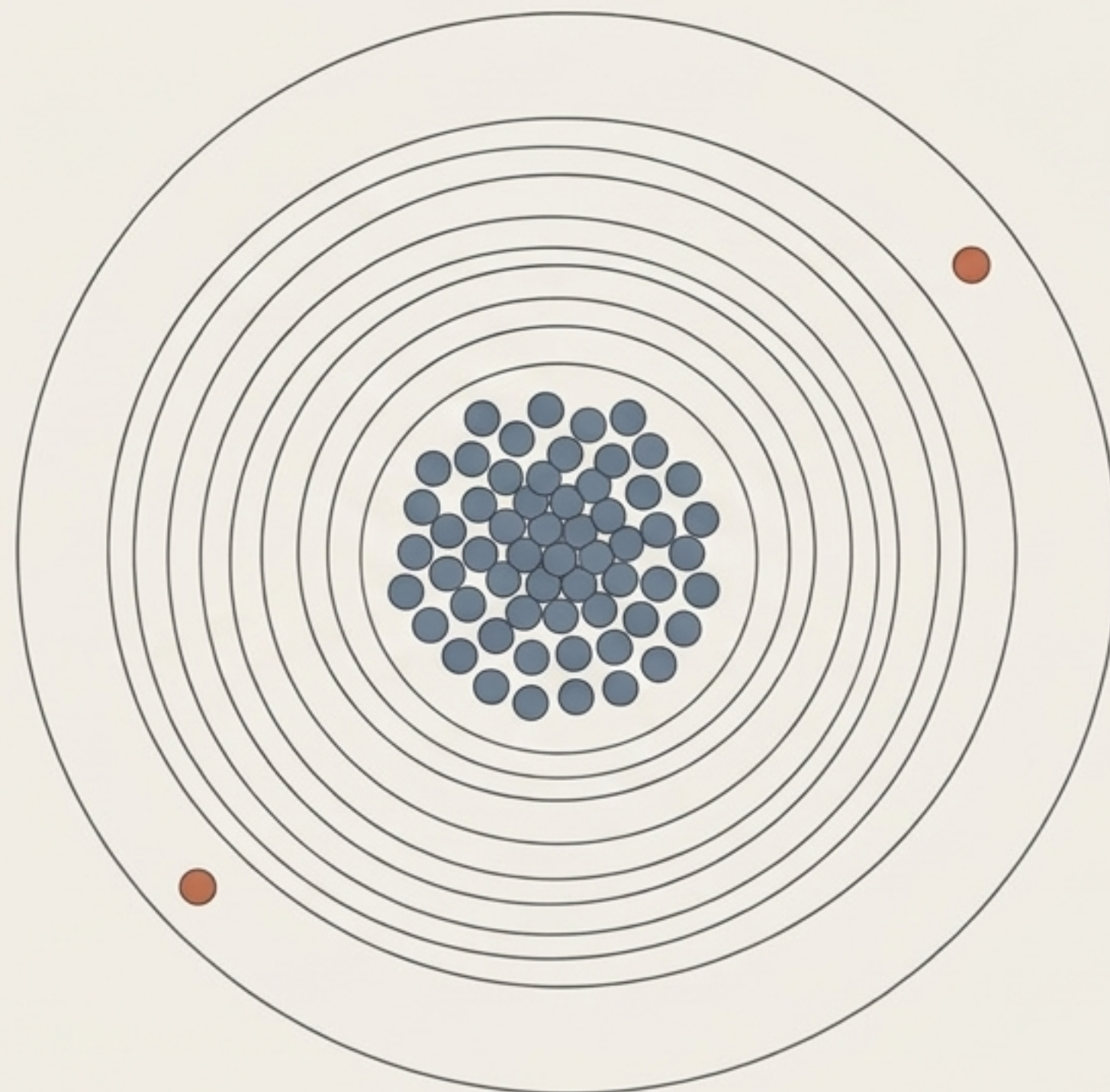
Hyperparameters: eps (default=0.5), min_samples (default=5).

```
dbscan = DBSCAN(eps=0.5, min_samples=5)
```

HDBSCAN

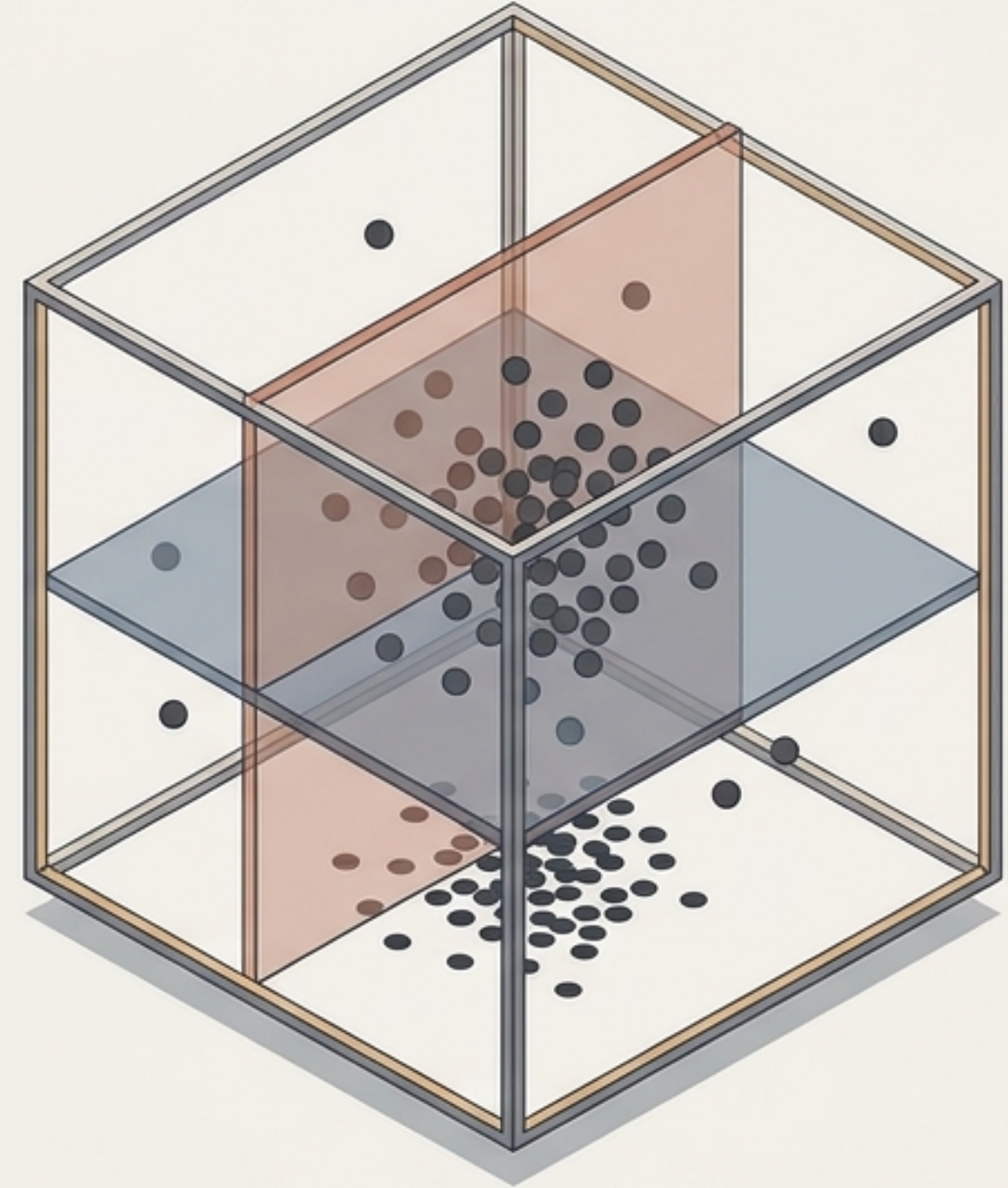
Mechanics: A variant of DBSCAN using cluster stability rather than distance parameters.

Advantage: Does not require setting specific distance parameters upfront.



Core Branch 2: The art of simplification

Dimensionality reduction simplifies data structure and reduces noise while retaining the maximum possible variance. It is crucial for feature extraction, data visualization, and improving clustering outcomes.



Reduction playbook: Linear simplification via PCA

Mechanics: A linear method that reduces dimensionality and noise by finding axes of maximum variance.

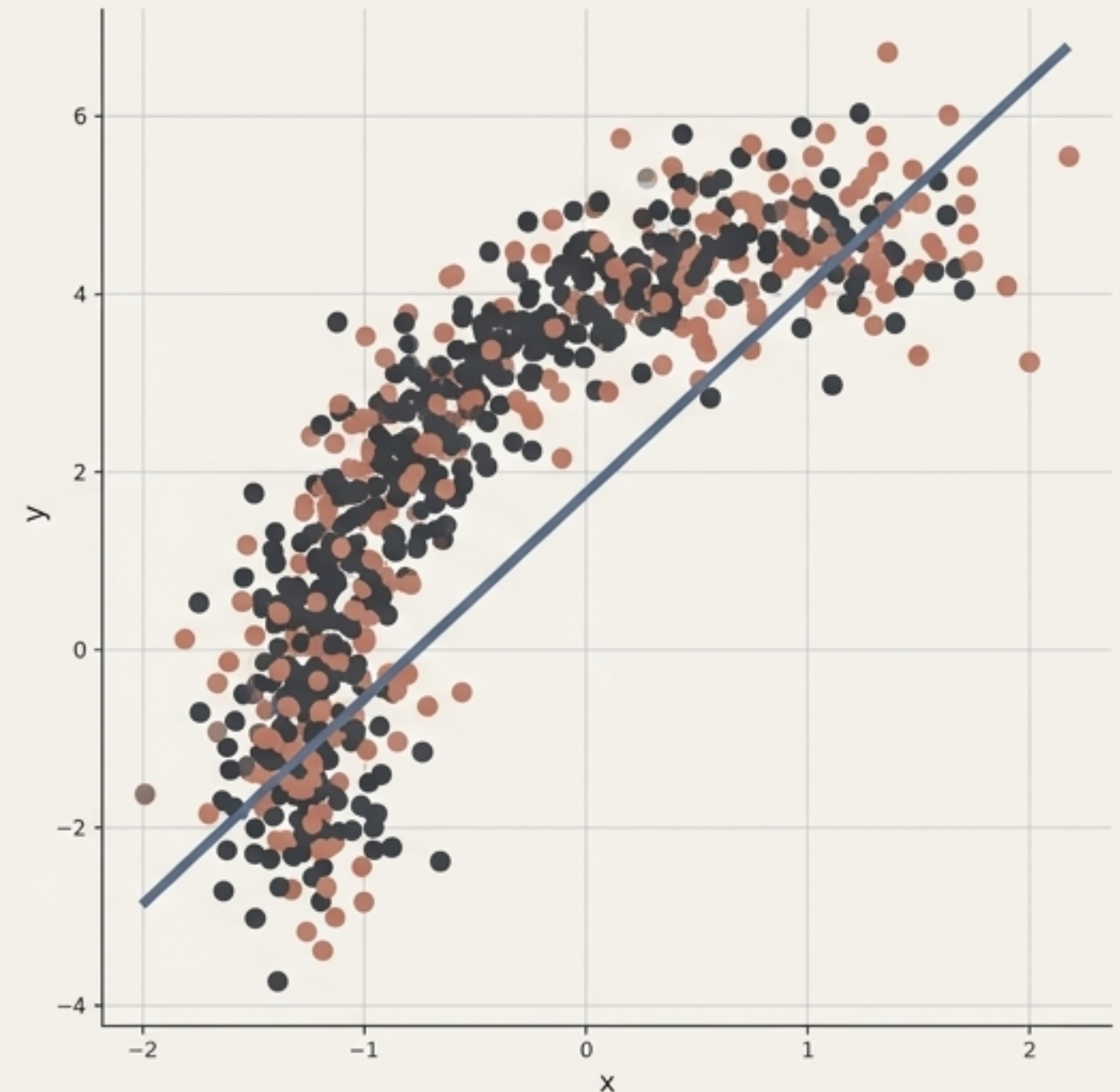
Pros/Cons: Easy to interpret, effectively reduces noise. Loses information in highly nonlinear data.

Hyperparameters: `n_components` (default=2), `whiten` (default=False), `svd_solver` (default='auto').

Pro-Tip

Use `pca.explained_variance_ratio_` to return the proportion of variance explained by each principal component.

```
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
```



Reduction playbook: Nonlinear mapping for complex data

t-SNE

t-Distributed Stochastic Neighbor Embedding

Mechanics: Preserves local similarities (neighbors) to visualize complex high-dimensional data.

Cons: Computationally expensive, sensitive to hyperparameters, prone to overfitting.

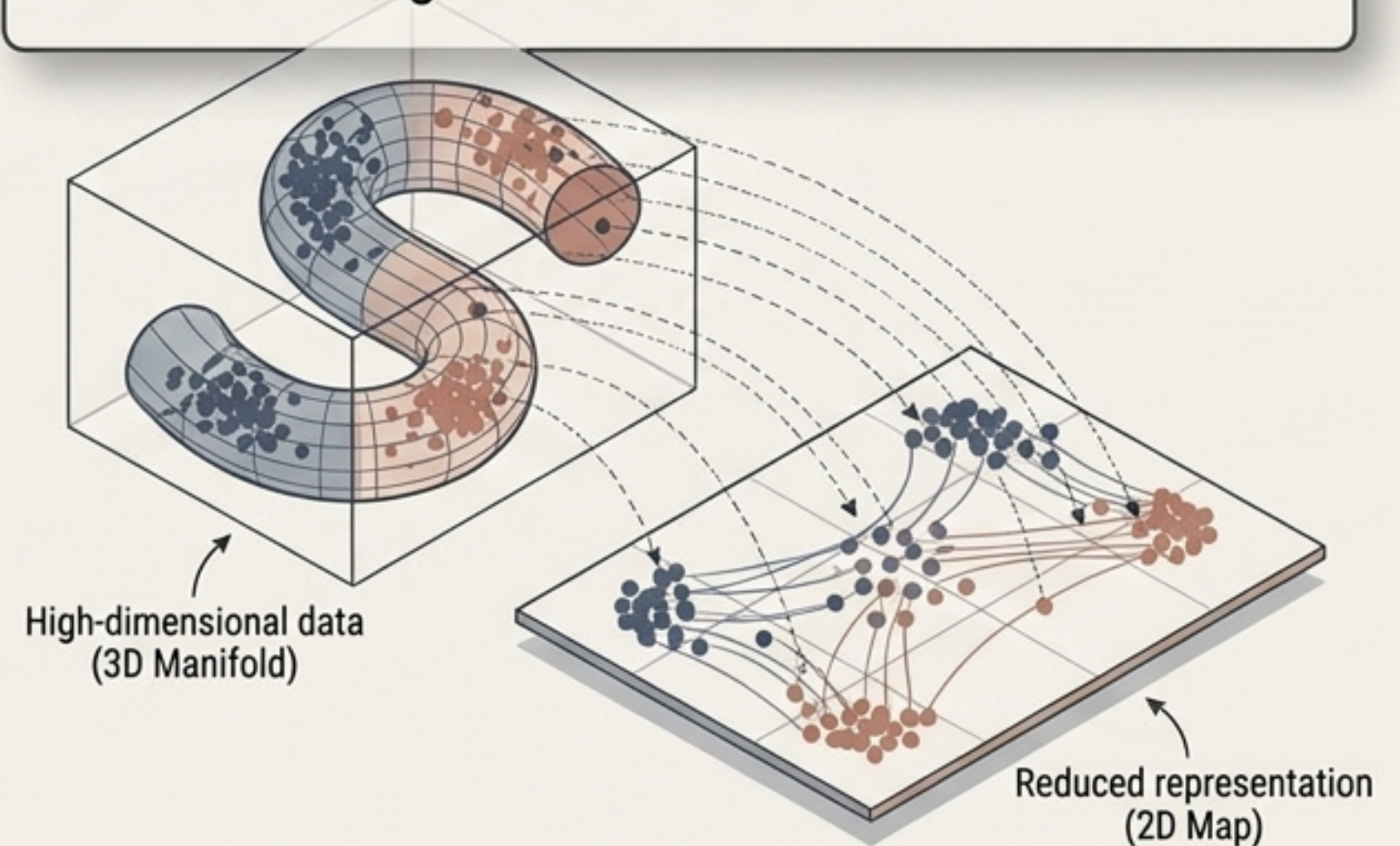
Hyperparameters: n_components (default=2), perplexity (default=30), learning_rate (default=200).

```
tsne = TSNE(n_components=2, perplexity=30, learning_rate=200)
```

UMAP

Uniform Manifold Approximation and Projection

Mechanics: Based on manifold theory. Preserves both local and global data structures.



Applied technique: Catching anomalies in the wild

Anomaly detection identifies rare data points that deviate significantly from the rest (e.g., fraud detection). By using unsupervised methods, you can catch unusual patterns before applying other machine learning algorithms.

1.

Prep your data

Scaling is required for DBSCAN, but is not required for Isolation Forests.

2.

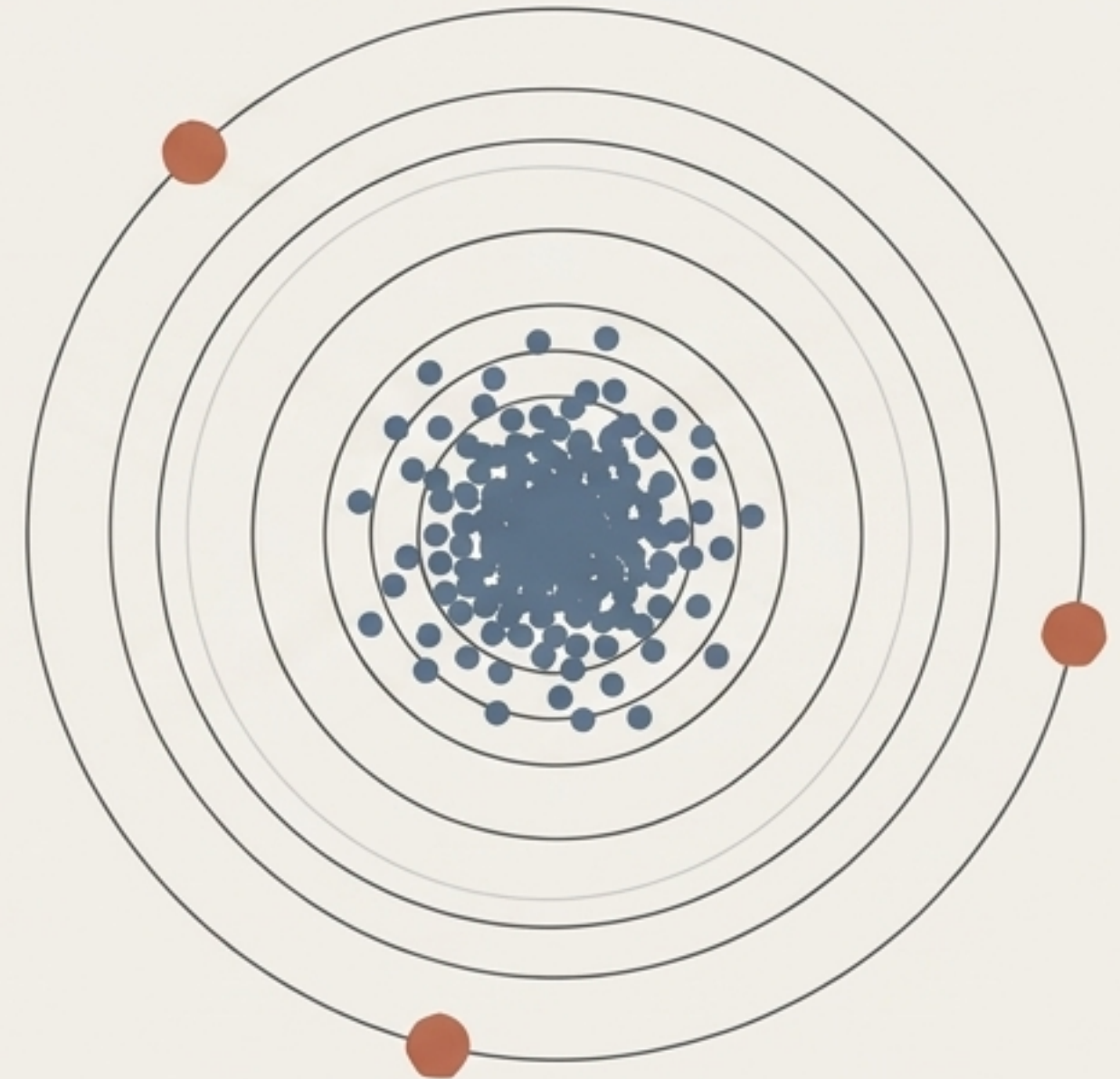
Start with Isolation Forests

Implement, visualize, tune, and select the model.

3.

Try DBSCAN

Compare the density-based results, visualize, tune, and select the best fit.



Applied technique: Powering recommendation engines

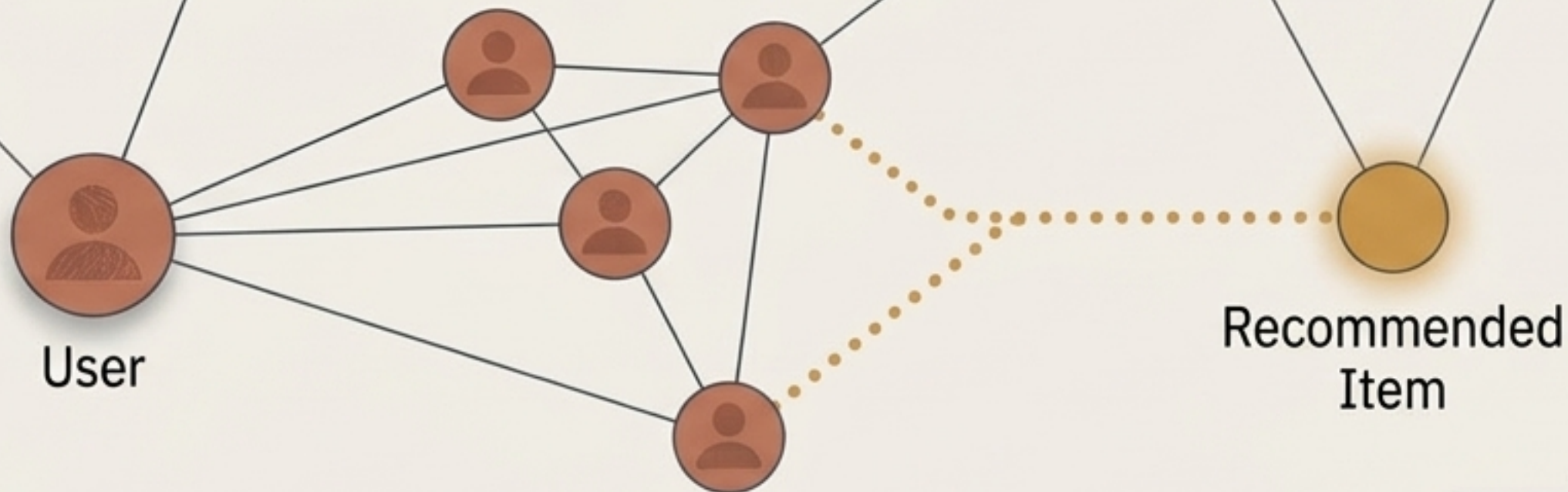
Suggesting items to users based on hidden behavioral similarities. Recommenders combine multiple machine learning techniques to map user preferences.

Unsupervised Core

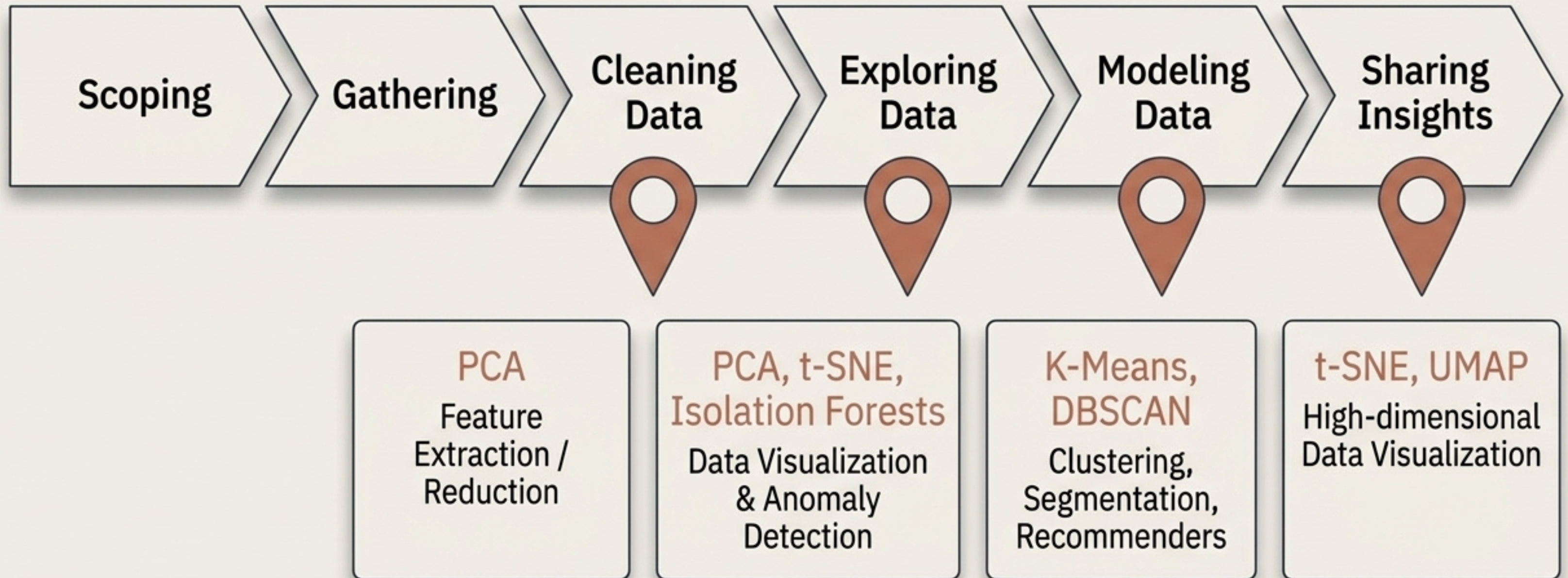
Relies heavily on Clustering (grouping similar users) and Dimensionality Reduction (simplifying complex user-item matrices).

Supporting Techniques

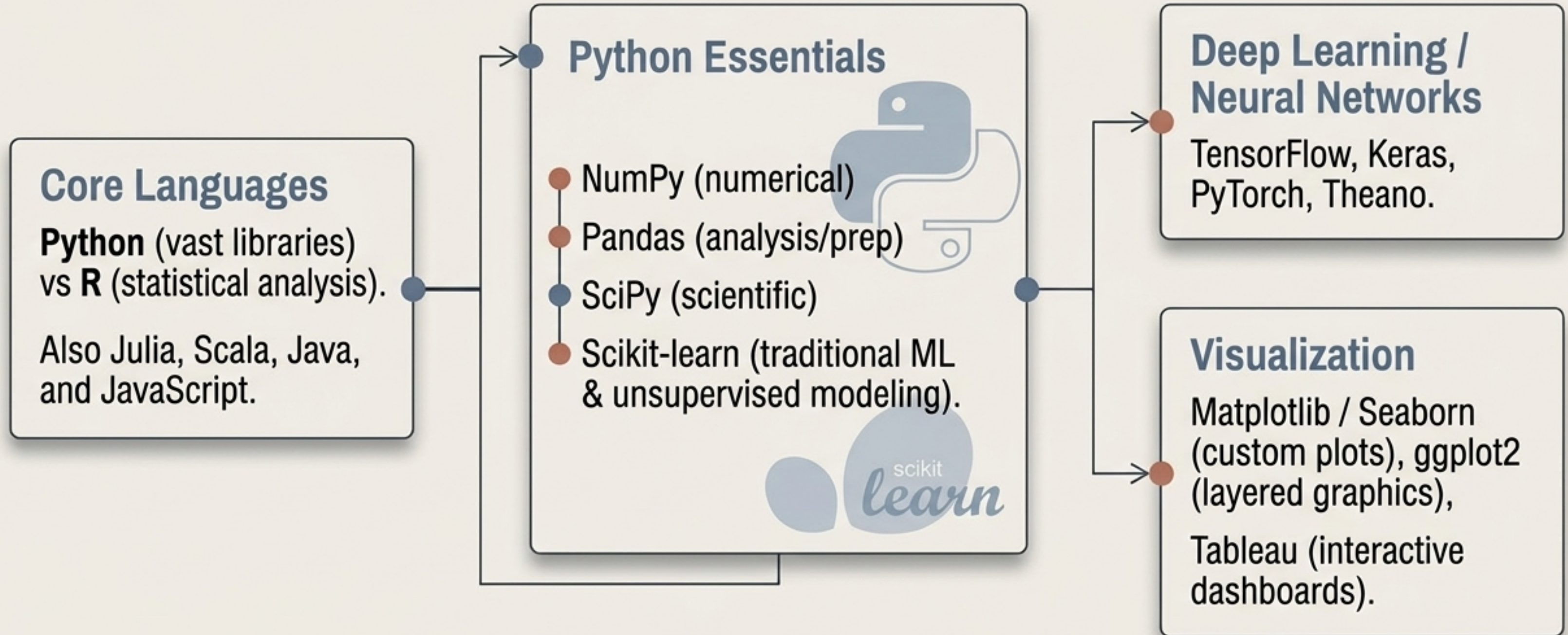
Recommenders also utilize Statistical Analysis, Distance Measures, and Time Series Analysis.



Unsupervised learning accelerates every phase of the data science workflow



The machine learning tooling ecosystem



The Unsupervised Playbook

Seek **structure**, not predictions.

Unsupervised learning operates without labels. The goal is to describe non-obvious organization in raw data.

Intuition drives evaluation.

Unlike supervised learning which relies on mathematical accuracy metrics, unsupervised evaluation requires heavy domain expertise and business **intuition**.

Group or Reduce.

The discipline splits into two primary actions: **Clustering** to find similarities, or Dimensionality Reduction to simplify complexity.

A **full-lifecycle** utility.

It is not just a modeling endpoint. Unsupervised techniques clean, explore, and visualize data throughout the entire analytical pipeline.

